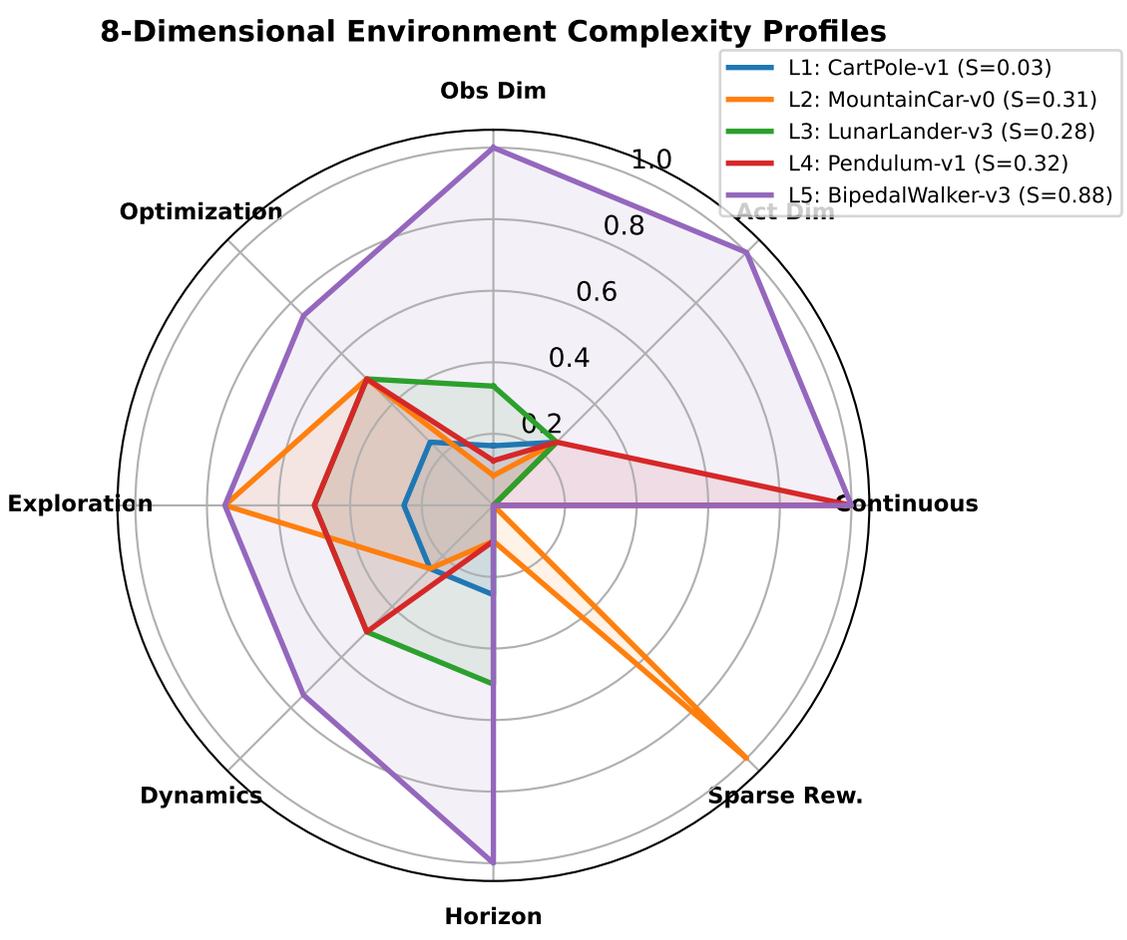
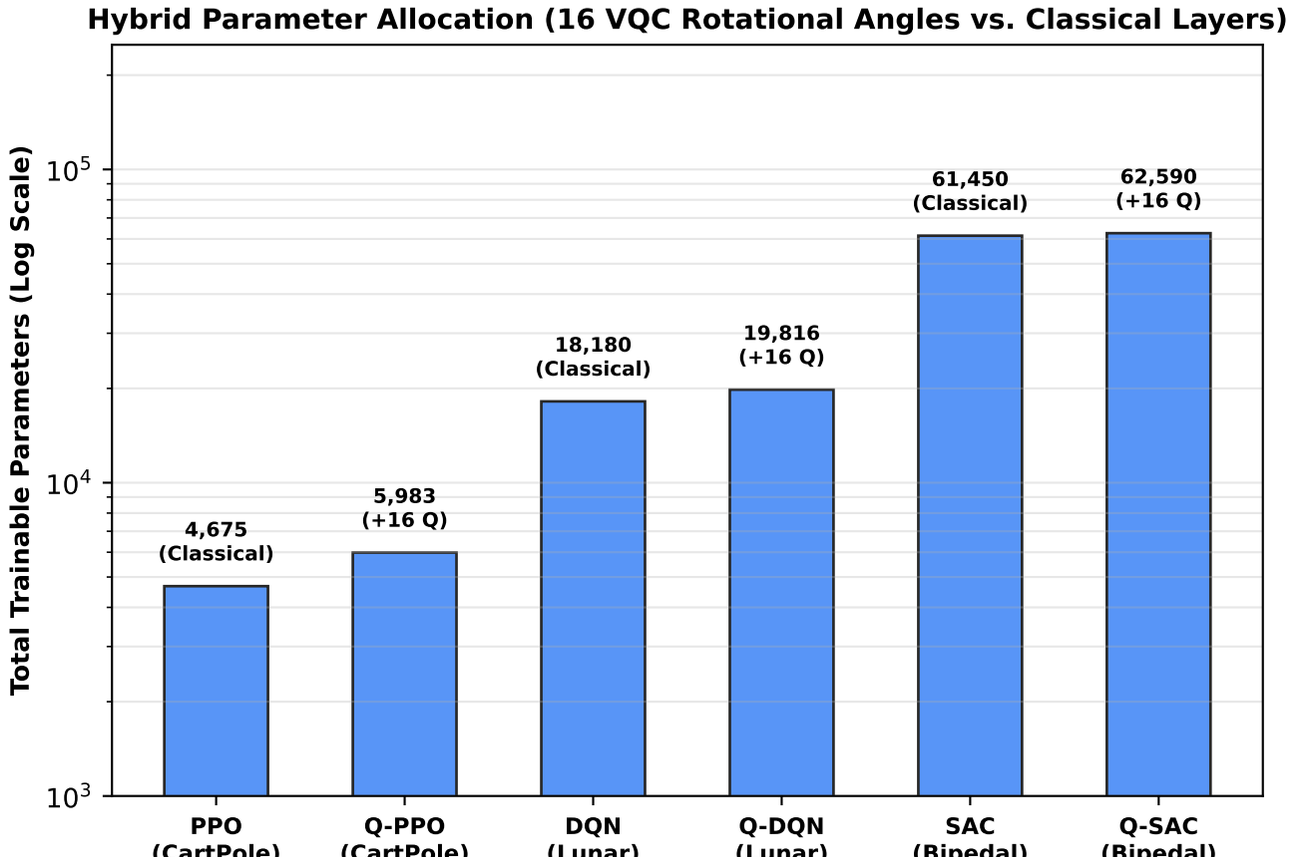

8-Dimensional Environment Complexity Profiles
L1: CartPole-v1 (S=0.03)
L2: MountainCar-v0 (S=0.31)
L3: LunarLander-v3 (S=0.28)
L4: Pendulum-v1 (S=0.32)
L5: BipedalWalker-v3 (S=0.88)
Obs Dim
Act Dim
Optimization
Exploration
Continuous
Dynamics
Sparse Rew.
Horizon
1.0
0.8
0.6
0.4
0.2


Hybrid Parameter Allocation (16 VQC Rotational Angles vs. Classical Layers)
Total Trainable Parameters (Log Scale)
10^5
10^4
10^3
4,675
(Classical)
5,983
(+16 Q)
18,180
(Classical)
19,816
(+16 Q)
61,450
(Classical)
62,590
(+16 Q)
PPO
(CartPole)
Q-PPO
(CartPole)
DQN
(Lunar)
Q-DQN
(Lunar)
SAC
(Bipedal)
Q-SAC
(Bipedal)